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DC Motor Vibration Survey & Fault Frequency Worksheet

For maintenance and reliability engineers surveying PMDC, BLDC and geared motors up to 15 kW

Use this worksheet at commissioning to record a baseline, and again whenever vibration is suspected. Compare like with like: same positions, speed, load and temperature.

Step 1 — Record operating conditions

- [ ] Record motor nameplate data, speed in rpm, load and frame temperature at the time of measurement.
- [ ] Calculate running-speed frequency: fr = rpm / 60 (3,000 rpm → 50 Hz).
- [ ] Record mounting type (B3 foot, B5 flange, B14 face) and drive type (coupling, belt, gearbox).
- [ ] Note any recent work: rewind, bearing change, coupling change or relocation.

Step 2 — Take broadband readings

- [ ] Mount the accelerometer with a magnet on clean, flat, bare metal on each bearing housing or end shield.
- [ ] Take velocity RMS readings over 10–1,000 Hz at DE and NDE in horizontal, vertical and axial directions (six readings).
- [ ] Classify the highest reading: Zone A ≤ 0.71 mm/s, Zone B 0.71–1.8 mm/s, Zone C 1.8–4.5 mm/s, Zone D > 4.5 mm/s (ISO 10816-1 Class I guidance).
- [ ] Capture a velocity spectrum to at least 10 × fr and an acceleration or envelope spectrum to a few kHz.
- [ ] Never mount the sensor on thin covers or fan cowls.

Step 3 — Separation tests

- [ ] Power-cut test: cut supply at speed. Instant disappearance → current-driven source (ripple, commutation, PWM). Gradual decay → mechanical source. Cogging in PM motors persists while coasting.
- [ ] Uncoupled test: run the motor alone. Vibration falls to bench level → coupling, load or base. Remains high → motor.
- [ ] Speed sweep: if vibration peaks in one narrow speed band only, suspect resonance and perform a bump test.
- [ ] Bolt test: slacken one hold-down bolt at a time; a large change in reading indicates soft foot or looseness.

Step 4 — Calculate fault frequencies

- [ ] Unbalance: 1 × fr, radial, steady phase. Force $F = m \times e \times \omega^2$, rising with speed squared.
- [ ] Misalignment: 2 × fr radial and 1 × fr axial.
- [ ] Bearing outer race: $BPFO = (n/2) \times fr \times (1 - (d/D) \cos \varphi)$. Inner race: $BPFI = (n/2) \times fr \times (1 + (d/D) \cos \varphi)$.
- [ ] Commutator bar passing: f = number of bars × rpm / 60.
- [ ] Cogging: f = LCM(slots, poles) × rpm / 60.
- [ ] Rectifier ripple on 50 Hz single-phase full-wave supply: 100 Hz.

Step 5 — Mounting and alignment checks (isolate first)

- [ ] Soft foot: dial indicator on each foot while slackening its bolt; investigate lift above about 0.05 mm.
- [ ] Alignment: laser or reverse dial method; target about 0.05 mm offset.
- [ ] Check coupling condition, pulley position and belt tension.

- [ ] Check commutator runout with a dial indicator against the manufacturer's limit.
- [ ] Check shaft end play and bearing fit in the end shields.

Step 6 — Correct and verify

- [ ] Correct one cause at a time and repeat all six readings under the same conditions.
- [ ] Balance rewound armatures dynamically before refitting and request the balancing report.
- [ ] Record the corrected readings as the new baseline and set a re-survey interval.
- [ ] Investigate any sustained step change from baseline, even within Zone B.

Fault Frequency Quick Reference

Dominant Frequency	Likely Cause	Confirming Check
1 × fr radial, steady	Unbalance	Rises with speed squared; persists uncoupled
1 × fr axial + 2 × fr radial	Misalignment	Alignment check; warm coupling
1 × fr, bolt-sensitive	Soft foot	Dial indicator on each foot
Many harmonics, 0.5 × fr	Rotating looseness	Unsteady phase; shaft play
Non-synchronous peaks	Bearing defect	Envelope spectrum at BPFO or BPFI
Bars × rpm / 60	Commutator runout or high bar	Runout check; sparking
One speed band only	Resonance	Bump test
100 Hz on mains supply	Rectifier ripple	Gone on battery or power cut

Zone limits are ISO 10816-1 Class I guidance values for installed machines up to 15 kW (ISO 10816-1 has since been revised as ISO 20816-1). Factory acceptance limits under IEC 60034-14 apply to uncoupled motors on the test bench. Confirm specific limits with the motor manufacturer.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.